

OLIST E-COMMERCE DATASET: TECHNICAL REPORT

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1. Project Overview

This portfolio project is an end-to-end data analytics pipeline built on the Olist Brazilian E-Commerce dataset. It covers raw data ingestion, schema validation, multi stage data cleaning, data analysis via SQL and interactive data visualization in Power BI.

URL: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

2. Dataset Description

The data originates from the Kaggle "Brazilian E-Commerce Public Dataset by Olist". The dataset consists of 8 relational CSV files representing operations between late 2016 and late 2018 (edge months have excluded in the final analysis due to incomplete volume).

The core tables include ~100,000 orders, ~96,000 customers, ~3,000 sellers, ~33,000 products and over 1,000,000 raw geolocation records.

3. Objective

To create a data pipeline that transforms messy, real-world data into actionable business insights to mitigate logistics bottlenecks and improve product quality.

4. Scope

Database schema design, staging table creation (`\_staging\_1`, `\_staging\_2`), data type casting, orphan checks, duplicate resolution, SQL CTE-driven analysis and Power BI dashboard (connected database via a read-only MySQL user).

5. Why a Business-Focused Angle?

Portfolio projects often limited to generic descriptive analytics (e.g., total sales, demographics) and focused on advanced technical concepts. I chose an operational and business focused angle because metrics like "gross revenue" are vanity metrics if separated from the cost of fulfillment and customer churn. Therefore, I focused to finding leverages that may improve customer satisfaction and increase customer retention.

The ultimate objective is to replicate actual work of a data analyst that is find gaps in current systems, leverage that to reduce costs and improve performance, rather than just reporting historical totals.

6. Methodology and Approach

6.1. Import & Validation:

Loaded raw CSV files into MySQL using `'LOAD DATA LOCAL INFILE'` (Preferred local data loading to increase importing speed). Validated baseline row counts explicitly to prevent silent data loss during load.

6.2. Multi Stage Cleaning:

Utilized staging tables to isolate transformations. Applied `'TRIM'` and `'NULLIF'` to clean strings and `'STR_TO_DATE'` for safe VARCHAR to DATETIME casting.

6.3. Data Integrity Check:

Conducted orphan checks using `'LEFT JOIN'` logic to ensure all foreign keys (e.g., `'order_id'` in reviews, `'customer_id'` in orders) existed in their parent tables. Added Primary Keys only after proving column uniqueness.

6.4. Data Modeling:

Used Common Table Expressions (CTEs) and ranking functions to aggregate data before final `'SELECT'` statements, optimizing query readability and performance.

7. Major Challenges and Resolutions

7.1. Geolocation Data Fan-Out.

The geolocation table contained multiple conflicting latitude/longitude coordinates per `'zip_code_prefix'` (over 980,000 duplicates), which would cause massive table duplication on joins.

Resolution: Implemented a `'ROW_NUMBER () OVER (PARTITION BY geolocation_zip_code_prefix ORDER BY geolocation_lat)'` window function to deterministically select a single coordinate per zip code, reducing the table to 19,015 clean records.

7.2. Many to Many Reviews and Orders.

Data profiling revealed 789 reviews mapped to multiple orders and 547 orders linked to multiple reviews. Deleting these as duplicates would erase valid data such as updated user reviews.

Resolution: Intentionally omitted a Primary Key on the staging table to preserve data. Handled the fan out downstream during analysis using a `'latest_review'` CTE that partitioned by `'order_id'` and sorted by `'review_creation_date DESC'` to isolate the most relevant score.

7.3. Silent Data Truncation/Loss.

Converting text timestamps to DATETIME risks silently converting invalid formats to NULL.

Resolution: Created temporary `\_2` columns, ran `STR\_TO\_DATE` and explicitly queried to ensure the `COUNT ()` of valid text dates perfectly matched the `COUNT ()` of successful DATETIME conversions before dropping the original columns.

8. Results

8.1. Logistics Bottlenecks:

Late delivery is highly regional. States like Alagoas (24% late) and Maranhao (20% late) in the North/Northeast severely lag behind the national average. Furthermore, the `carrier\_to\_customer` leg accounts for the vast majority of delays, proving that seller dispatch times are not the primary issue.

8.2. Capacity Scaling Failures:

From late 2017 to Q1 2018, revenue and order volumes peaked. However, the `late\_delivery\_pct` increased and `avg\_review\_score` dropped inversely during this exact window, highlighting an operations capacity failure under load.

8.3. Product Category Gaps:

Using a combined rank query (revenue rank + low score rank), the analysis identified 'Bed Bath Table', 'Office Furniture' and 'Watches Gifts' as categories generating massive revenue but suffering from disproportionately low customer satisfaction scores.

9. Conclusions

The data indicate inefficiency in logistics strategy. Logistics optimizations (carrier SLA renegotiations or routing overhauls) must be targeted on last mile carrier performance in the North/Northeast regions.

Additionally, because revenue growth historically dropped delivery capacity, predictive supply chain scaling is required ahead of Q4 volumes.

Finally, quality assurance investments should be directed immediately at the 'Bed Bath Table' and 'Watches Gifts' categories, as their high volume means incremental quality improvements will yield the highest mathematical impact on overall platform retention.